

```
[2]: import numpy as np
import pandas as pd
```

```
[3]: data = pd.read_csv('/home/subramanyaprasad/Machine Learning/Data Analysis/Python Data Science Handbook/Files/president_heights.csv')
```

```
[24]: data
```

	order	name	height(cm)
0	1	George Washington	189
1	2	John Adams	170
2	3	Thomas Jefferson	189
3	4	James Madison	163
4	5	James Monroe	183
5	6	John Quincy Adams	171
6	7	Andrew Jackson	185
7	8	Martin Van Buren	168
8	9	William Henry Harrison	173
9	10	John Tyler	183
10	11	James K. Polk	173
11	12	Zachary Taylor	173
12	13	Millard Fillmore	175
13	14	Franklin Pierce	178
14	15	James Buchanan	183
15	16	Abraham Lincoln	193
16	17	Andrew Johnson	178
17	18	Ulysses S. Grant	173
18	19	Rutherford B. Hayes	174
19	20	James A. Garfield	183
20	21	Chester A. Arthur	183
21	23	Benjamin Harrison	168
22	25	William McKinley	170
23	26	Theodore Roosevelt	178
24	27	William Howard Taft	182
25	28	Woodrow Wilson	180
26	29	Warren G. Harding	183
27	30	Calvin Coolidge	178
28	31	Herbert Hoover	182
29	32	Franklin D. Roosevelt	188
30	33	Harry S. Truman	175
31	34	Dwight D. Eisenhower	179
32	35	John F. Kennedy	183
33	36	Lyndon B. Johnson	193
34	37	Richard Nixon	182
35	38	Gerald Ford	183
36	39	Jimmy Carter	177
37	40	Ronald Reagan	185
38	41	George H. W. Bush	188
39	42	Bill Clinton	188
40	43	George W. Bush	182
41	44	Barack Obama	185
42	45	Donald Trump	191
43	46	Joseph Biden	182

```
[10]: heights = np.array(data['height(cm)'])
print(heights)

[189 170 189 163 183 171 185 168 173 183 173 173 175 178 183 193 178 173
 174 183 183 168 170 178 182 180 183 178 182 188 175 179 183 193 182 183
 177 185 188 188 182 185 191 182]
```

Average height:

```
[11]: print("Average height: ",heights.mean())

Average height:  180.04545454545453

Median height:
```

```
[14]: print("Median height: ",np.median(heights))

Median height:  182.0

Standard deviation:
```

```
[15]: print("Standard deviation: ",heights.std())

Standard deviation:  6.983599441335736

Min and max heights:
```

```
[16]: print("Minimum height: ",heights.min())
print("Maximum height: ",heights.max())

Minimum height:  163
Maximum height:  193

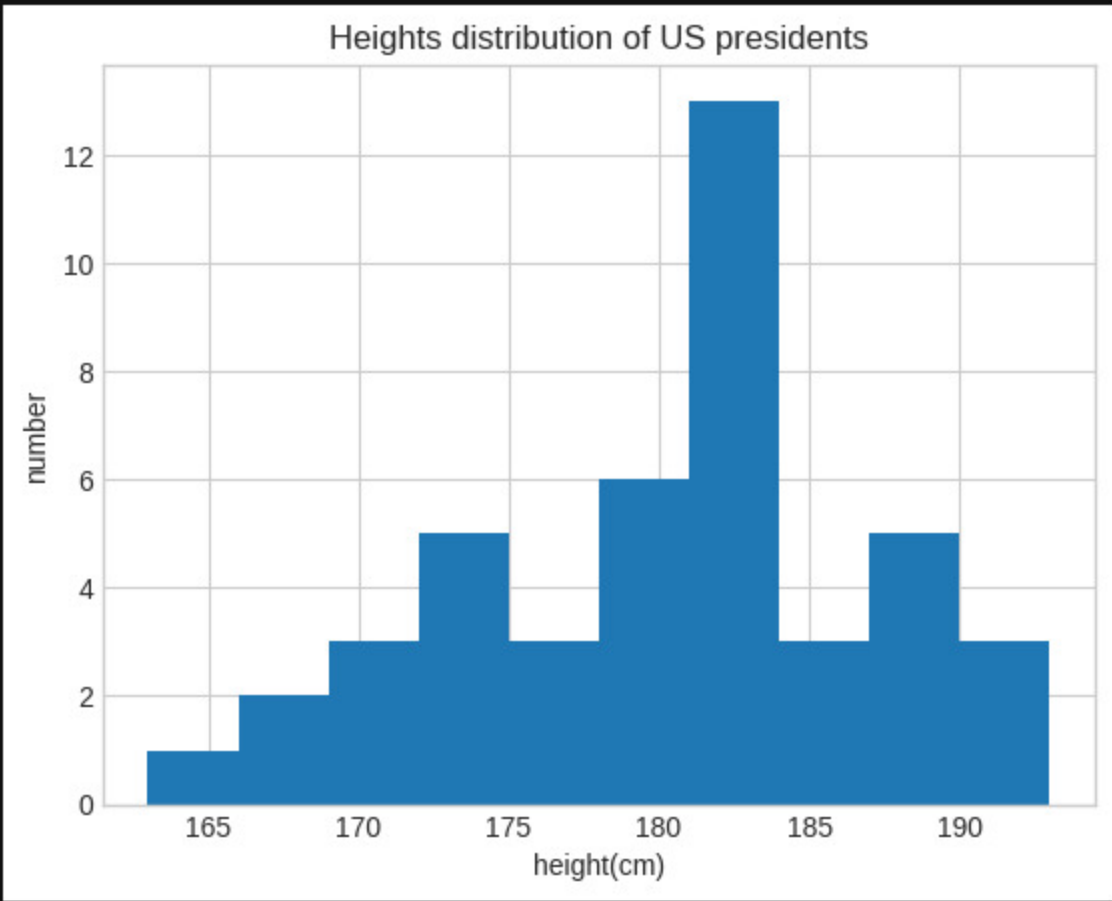
Visualization:
```

```
[17]: import matplotlib.pyplot as plt
```

```
[21]: plt.style.use('seaborn-v0.8-whitegrid')
```

```
[23]: plt.hist(heights)
plt.title("Heights distribution of US presidents")
plt.xlabel('height(cm)')
plt.ylabel('number')
```

```
[23]: Text(0, 0.5, 'number')
```



```
[ ]:
```